

NAT and PAT -- Guided Notes ANSWER KEY

Unit 2.3.2 | CompTIA Network+ N10-009 Obj. 2.3 | N10-008 2.3

For instructor use / distribute after students complete the notes.

Question 1

NAT stands for _____. It was developed to extend the life of IPv4 by allowing _____ of devices to share a single _____ IP address, delaying the exhaustion of the public address pool.

Answer: Network Address Translation; many (an entire network); public.

Question 2

The three private (non-routable) IPv4 address ranges are _____, _____, and _____. Packets with these source addresses are _____ by internet backbone routers.

Answer: 10.0.0.0/8; 172.16.0.0/12; 192.168.0.0/16; dropped (not forwarded).

Question 3

Static NAT creates a _____ -to- _____ permanent mapping between a private address and a public address. This is typically used for devices that must be _____ from the internet, such as web servers or mail servers.

Answer: One-to-one; permanent; reachable (accessible).

Question 4

PAT (Port Address Translation) is also called NAT _____. It allows thousands of devices to share a _____ public IP address by using unique _____ numbers to track each session in the NAT translation table.

Answer: Overload; single; port.

Question 5

In NAT terminology, the _____ interface faces the private network, while the _____ interface faces the public internet. When a packet leaves the private network, NAT replaces the _____ address with the router's public IP.

Answer: Inside (or inside local); outside (or inside global); source (private).

Question 6

Describe one advantage and one disadvantage of NAT. How does NAT affect end-to-end connectivity, and why is this a problem for certain protocols or applications?

Answer: Advantage: conserves public IPv4 addresses; also provides a degree of security by hiding internal addressing. Disadvantage: breaks end-to-end connectivity -- devices behind NAT cannot be directly addressed from outside, which complicates peer-to-peer applications, VoIP, and some VPN protocols that embed IP addresses in the payload.

Question 7

Dynamic NAT maps private addresses to a _____ of public addresses. Unlike PAT, each active session uses a _____ public IP. If the pool is exhausted, new connections are _____ until a public address becomes available.

Answer: Pool; dedicated (unique); dropped (queued/refused).

Question 8 -- Real World Application

REAL WORLD: A user reports they can access the internet normally, but a colleague trying to connect to the user's PC from outside the office cannot reach it. The office uses PAT on a single public IP. Explain why this happens and describe two solutions that would allow inbound connections.

Answer: PAT only tracks outbound sessions -- unsolicited inbound connections have no NAT table entry, so the router drops them. Solutions: (1) Port forwarding -- configure a static mapping from a specific public port to the internal IP:port of the target PC; (2) Static NAT -- assign the PC a dedicated public IP. A VPN where the colleague initiates the tunnel outbound is a third option.